In [21]:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, f1_score, recall_score
from scipy.optimize import minimize
```

In [22]:

```
data = [[ 2.0, 2.2 , 1],
        [2.7 ,2.5 , 1],
        [2.3, 2.0 , 1],
        [3.1, 2.3 , 1],
        [2.5 ,2.4 , 1],
        [2.8 ,2.7 , 1],
        [1.6, 1.5 ,-1],
        [2.0 ,1.9 ,-1],
        [2.1, 1.8 ,-1],
        [1.7, 1.6, -1],
        [1.8 ,1.7 ,-1],
        [2.0, 1.6 ,-1]]
```

In [23]:

```
df = pd.DataFrame(data)
```

In [24]:

```
df.columns = ["x1", "x2", "y"]
```

In [25]:

```
df.head(5)
```

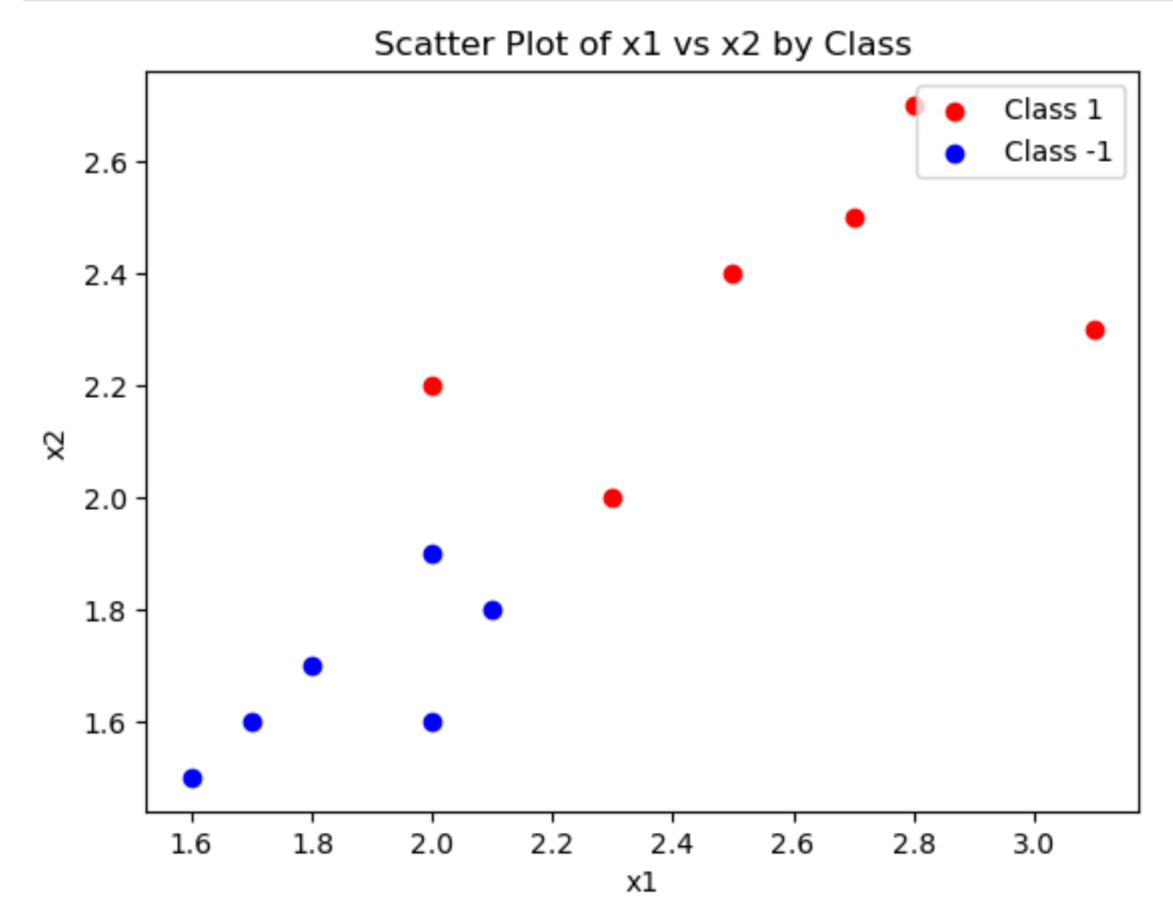
Out[25]:

| | x1 | x2 | y |
|---|---|---|---|
| **0** | 2.0 | 2.2 | 1 |
| **1** | 2.7 | 2.5 | 1 |
| **2** | 2.3 | 2.0 | 1 |
| **3** | 3.1 | 2.3 | 1 |
| **4** | 2.5 | 2.4 | 1 |

In [26]:

```
# Plot
plt.scatter(df[df['y'] == 1]['x1'], df[df['y'] == 1]['x2'], color='red', label='Class 1')
plt.scatter(df[df['y'] == -1]['x1'], df[df['y'] == -1]['x2'], color='blue', label='Class -1')

# Add labels and legend
plt.xlabel('x1')
plt.ylabel('x2')
plt.title('Scatter Plot of x1 vs x2 by Class')
plt.legend()
plt.show()
```



In [27]:

```
X = np.array(df.iloc[:, : -1])
y = np.array(df.iloc[:, -1:])
```

In [28]:

```
def learn_svm(X, y):
    N, d = X.shape

    # Define C, c, A, a, B, b
    C = np.outer(y, y) * (X @ X.T)
    C += 1e-8 * np.eye(N)  # Regularization for stability
    c = -np.ones(N)
    A = y.reshape(1, -1)
    a = np.array([0])
    B = -np.eye(N)
    b = np.zeros(N)

    # Compute N+ and N-
    N_plus = np.sum(y == 1)
    N_minus = N - N_plus

    # Initialize alpha
    alpha_init = np.where(y == 1, 1 / N_plus, 1 / N_minus)

    # Quadratic programming solver
    def objective(alpha):
        return 0.5 * alpha @ C @ alpha + c @ alpha

    def equality_constraint(alpha):
        return A @ alpha - a

    constraints = [
        {"type": "eq", "fun": equality_constraint},
        {"type": "ineq", "fun": lambda alpha: alpha}  # Ensure alpha >= 0
    ]

    result = minimize(
        fun=objective,
        x0=alpha_init,
        constraints=constraints,
        method='SLSQP'
    )
    alpha = result.x

    # Compute beta and beta0
    beta = (alpha * y.ravel()) @ X
    support_vectors = np.where(alpha > 1e-5)[0]
    beta_0 = np.mean(y[support_vectors] - X[support_vectors] @ beta)

    return beta_0, beta
```

In [29]:

```
beta_0, beta = learn_svm(X, y)
print(f"Beta_0 (bias): {beta_0}")
print(f"Beta (weights): {beta}")
```

```
Beta_0 (bias): -22.55561873982946
Beta (weights): [4.44446865 6.66667275]
```

```
/var/folders/4m/t9k_jsy176ngtyczt72zczj40000gn/T/ipykernel_53695/1608809079.py:32: DeprecationWarning: Use of `minimize` with `x0.ndim != 1` is deprecated. Currently, s
ingleton dimensions will be removed from `x0`, but an error will be raised in SciPy 1.11.0.
  result = minimize(
```

In [30]:

```
# Compute predictions using the SVM model
# Decision boundary: beta_0 + beta[0] * x1 + beta[1] * x2
y_pred = np.sign(beta_0 + X @ beta)

# Compute metrics
accuracy = accuracy_score(y, y_pred)
f1 = f1_score(y, y_pred)
recall = recall_score(y, y_pred)

# Report metrics
print(f"Accuracy: {accuracy:.2f}")
print(f"F1 Score: {f1:.2f}")
print(f"Recall: {recall:.2f}")
```

```
Accuracy: 1.00
F1 Score: 1.00
Recall: 1.00
```

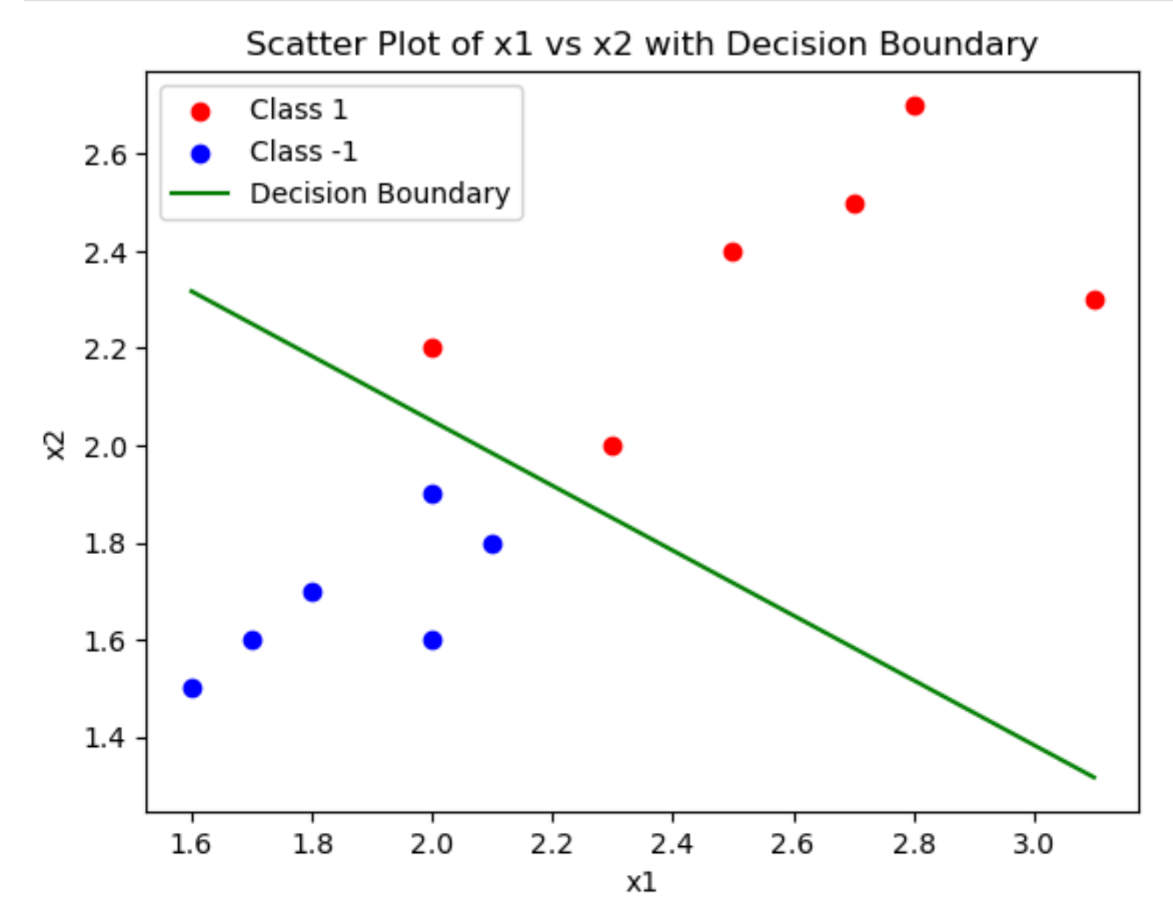
In [31]:

```
# Plot the data points
plt.scatter(df[df['y'] == 1]['x1'], df[df['y'] == 1]['x2'], color='red', label='Class 1')
plt.scatter(df[df['y'] == -1]['x1'], df[df['y'] == -1]['x2'], color='blue', label='Class -1')

# Compute the decision boundary
x1_vals = np.linspace(df['x1'].min(), df['x1'].max(), 100)
x2_vals = -(beta_0 + beta[0] * x1_vals) / beta[1]
plt.plot(x1_vals, x2_vals, color='green', label='Decision Boundary')

# Add labels and legend
plt.xlabel('x1')
plt.ylabel('x2')
plt.title('Scatter Plot of x1 vs x2 with Decision Boundary')
plt.legend()

plt.show()
```



In [ ]:

In [34]:

```
from imblearn.over_sampling import SMOTE
from imblearn.under_sampling import RandomUnderSampler
import torch
import torch.nn as nn
import torch.optim as optim

#loading the data
c_data = pd.read_csv('creditcard.csv', date_parser='Time')
#viewing the data
c_data.head(5)
```

Out[34]:

| | Time | V1 | V2 | V3 | V4 | V5 | V6 | V7 | V8 | V9 | ... | V21 | V22 | V23 | V24 | V25 | V26 | V27 | V28 | Amount | Class |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 0.0 | -1.359807 | -0.072781 | 2.536347 | 1.378155 | -0.338321 | 0.462388 | 0.239599 | 0.098698 | 0.363787 | ... | -0.018307 | 0.277838 | -0.110474 | 0.066928 | 0.128539 | -0.189115 | 0.133558 | -0.021053 | 149.62 | 0 |
| **1** | 0.0 | 1.191857 | 0.266151 | 0.166480 | 0.448154 | 0.060018 | -0.082361 | -0.078803 | 0.085102 | -0.255425 | ... | -0.225775 | -0.638672 | 0.101288 | -0.339846 | 0.167170 | 0.125895 | -0.008983 | 0.014724 | 2.69 | 0 |
| **2** | 1.0 | -1.358354 | -1.340163 | 1.773209 | 0.379780 | -0.503198 | 1.800499 | 0.791461 | 0.247676 | -1.514654 | ... | 0.247998 | 0.771679 | 0.909412 | -0.689281 | -0.327642 | -0.139097 | -0.055353 | -0.059752 | 378.66 | 0 |
| **3** | 1.0 | -0.966272 | -0.185226 | 1.792993 | -0.863291 | -0.010309 | 1.247203 | 0.237609 | 0.377436 | -1.387024 | ... | -0.108300 | 0.005274 | -0.190321 | -1.175575 | 0.647376 | -0.221929 | 0.062723 | 0.061458 | 123.50 | 0 |
| **4** | 2.0 | -1.158233 | 0.877737 | 1.548718 | 0.403034 | -0.407193 | 0.095921 | 0.592941 | -0.270533 | 0.817739 | ... | -0.009431 | 0.798278 | -0.137458 | 0.141267 | -0.206010 | 0.502292 | 0.219422 | 0.215153 | 69.99 | 0 |

5 rows × 31 columns

In [35]:

```
#analysing the data to understand its distribution
c_data.describe()
```

Out[35]:

| | Time | V1 | V2 | V3 | V4 | V5 | V6 | V7 | V8 | V9 | ... | V21 | V22 | V23 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 284807.000000 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | ... | 2.848070e+05 | 2.848070e+05 | 2.848070e+05 | 2.848 |
| **mean** | 94813.859575 | 3.918649e-15 | 5.682686e-16 | -8.761736e-15 | 2.811118e-15 | -1.552103e-15 | 2.040130e-15 | -1.698953e-15 | -1.893285e-16 | -3.147640e-15 | ... | 1.473120e-16 | 8.042109e-16 | 5.282512e-16 | 4.45 |
| **std** | 47488.145955 | 1.958696e+00 | 1.651309e+00 | 1.516255e+00 | 1.415869e+00 | 1.380247e+00 | 1.332271e+00 | 1.237094e+00 | 1.194353e+00 | 1.098632e+00 | ... | 7.345240e-01 | 7.257016e-01 | 6.244603e-01 | 6.05 |
| **min** | 0.000000 | -5.640751e+01 | -7.271573e+01 | -4.832559e+01 | -5.683171e+00 | -1.137433e+02 | -2.616051e+01 | -4.355724e+01 | -7.321672e+01 | -1.343407e+01 | ... | -3.483038e+01 | -1.093314e+01 | -4.480774e+01 | -2.836 |
| **25%** | 54201.500000 | -9.203734e-01 | -5.985499e-01 | -8.903648e-01 | -8.486401e-01 | -6.915971e-01 | -7.682956e-01 | -5.540759e-01 | -2.086297e-01 | -6.430976e-01 | ... | -2.283949e-01 | -5.423504e-01 | -1.618463e-01 | -3.54 |
| **50%** | 84692.000000 | 1.810880e-02 | 6.548556e-02 | 1.798463e-01 | -1.984653e-02 | -5.433583e-02 | -2.741871e-01 | 4.010308e-02 | 2.235804e-02 | -5.142873e-02 | ... | -2.945017e-02 | 6.781943e-03 | -1.119293e-02 | 4.09 |
| **75%** | 139320.500000 | 1.315642e+00 | 8.037239e-01 | 1.027196e+00 | 7.433413e-01 | 6.119264e-01 | 3.985649e-01 | 5.704361e-01 | 3.273459e-01 | 5.971390e-01 | ... | 1.863772e-01 | 5.285536e-01 | 1.476421e-01 | 4.39 |
| **max** | 172792.000000 | 2.454930e+00 | 2.205773e+01 | 9.382558e+00 | 1.687534e+01 | 3.480167e+01 | 7.330163e+01 | 1.205895e+02 | 2.000721e+01 | 1.559499e+01 | ... | 2.720284e+01 | 1.050309e+01 | 2.252841e+01 | 4.584 |

8 rows × 31 columns

In [38]:

```
# Check class distribution
print(c_data['Class'].value_counts())
```

```
0    284315
1       492
Name: Class, dtype: int64
```

In [39]:

```
# Split the dataset into features (X) and target (y)
X = c_data.drop(columns=['Class'])
y = c_data['Class']

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

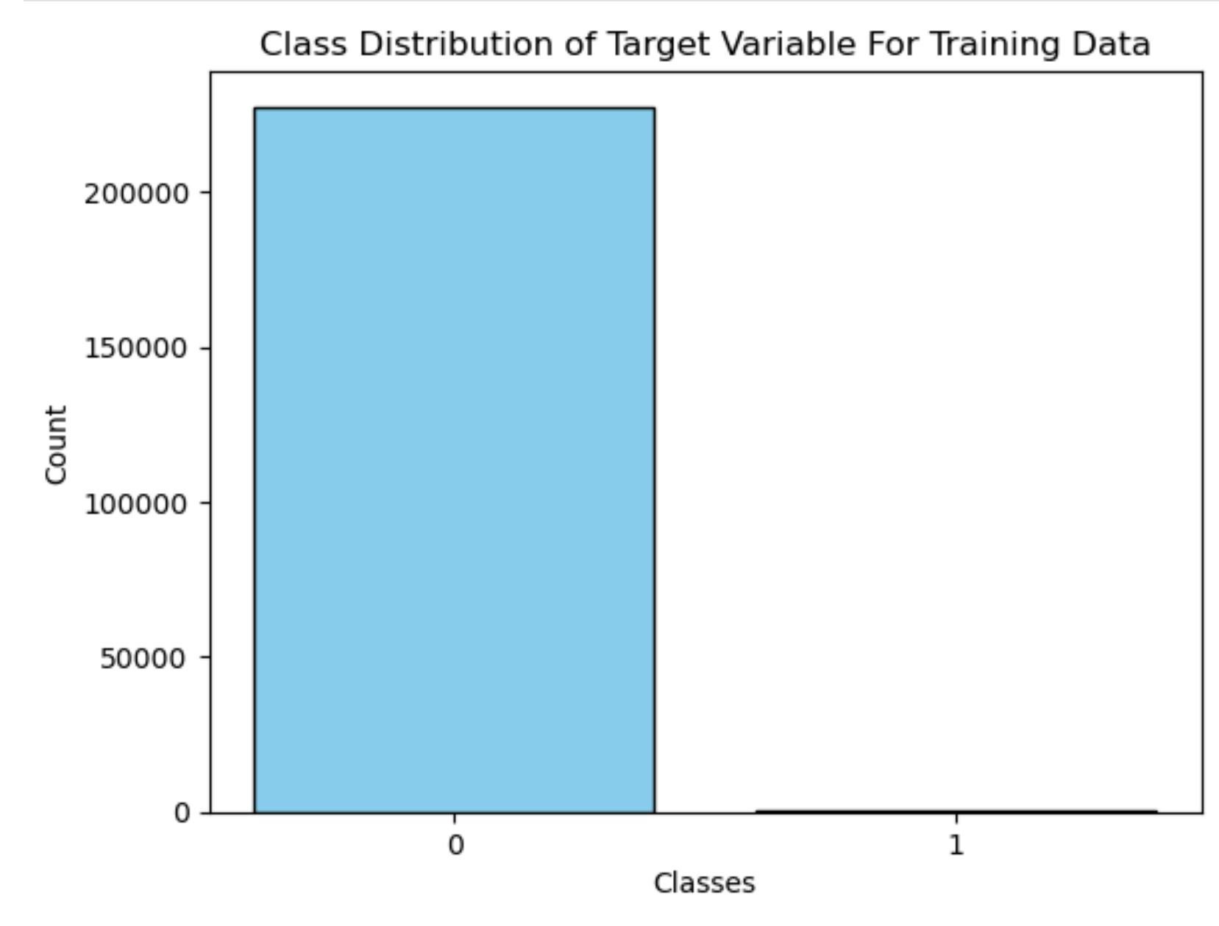
In [40]:

```
# Analyze class distribution in training  and test set
train_unique_classes, train_counts = np.unique(y_train, return_counts=True)
print(f"Class Distribution in Training Set: {dict(zip(train_unique_classes, train_counts))}")
```

```
Class Distribution in Training Set: {0: 227451, 1: 394}
```

In [41]:

```
# Plot the class distribution
plt.bar(train_unique_classes, train_counts, color='skyblue', edgecolor='black')
plt.title('Class Distribution of Target Variable For Training Data')
plt.xlabel('Classes')
plt.ylabel('Count')
plt.xticks(train_unique_classes)
plt.show()
```



In [42]:

```
# Apply SMOTE Oversampling
smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)

# Apply Random Undersampling
undersampler = RandomUnderSampler(random_state=42)
X_train_under, y_train_under = undersampler.fit_resample(X_train, y_train)

# Check distributions after resampling
print(f"SMOTE Resampled Training Set: {np.bincount(y_train_smote)}")
print(f"Undersampled Training Set: {np.bincount(y_train_under)}")
```

```
SMOTE Resampled Training Set: [227451 227451]
Undersampled Training Set: [394 394]
```

In [43]:

```
class MLP(nn.Module):
    def __init__(self, input_size):
        super(MLP, self).__init__()
        self.fc1 = nn.Linear(input_size, 128)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(128, 64)
        self.fc3 = nn.Linear(64, 1)  # Binary classification

    def forward(self, x):
        x = self.relu(self.fc1(x))
        x = self.relu(self.fc2(x))
        x = torch.sigmoid(self.fc3(x))  # Sigmoid activation for binary output
        return x
```

In [44]:

```
def train_model(X_train, y_train, model, criterion, optimizer, epochs=10):
    model.train()
    for epoch in range(epochs):
        # Convert data to tensors
        inputs = torch.tensor(X_train.values, dtype=torch.float32)
        labels = torch.tensor(y_train.values, dtype=torch.float32).unsqueeze(1)

        # Forward pass
        outputs = model(inputs)
        loss = criterion(outputs, labels)

        # Backward pass and optimization
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        print(f"Epoch {epoch+1}, Loss: {loss.item()}")
```

In [45]:

```
def evaluate_model(X_test, y_test, model):
    model.eval()
    with torch.no_grad():
        inputs = torch.tensor(X_test.values, dtype=torch.float32)
        labels = torch.tensor(y_test.values, dtype=torch.float32).unsqueeze(1)
        outputs = model(inputs).numpy()
        outputs = (outputs > 0.5).astype(int)  # Convert probabilities to binary predictions

        # Calculate metrics
        accuracy = accuracy_score(y_test, outputs)
        recall = recall_score(y_test, outputs)
        f1 = f1_score(y_test, outputs)

        print(f"Accuracy: {accuracy:.2f}")
        print(f"Recall: {recall:.2f}")
        print(f"F1-Score: {f1:.2f}")
```

In [46]:

```
# Original Dataset
input_size = X_train.shape[1]
model = MLP(input_size)
criterion = nn.BCELoss()  # Binary Cross-Entropy Loss for classification
optimizer = optim.Adam(model.parameters(), lr=0.001)

print("Training on Original Dataset:")
train_model(X_train, y_train, model, criterion, optimizer, epochs=10)
print("Evaluation on Original Dataset:")
evaluate_model(X_test, y_test, model)
```

```
Training on Original Dataset:
Epoch 1, Loss: 0.1858106106519699
Epoch 2, Loss: 0.17332890629768372
Epoch 3, Loss: 0.17310304939746857
Epoch 4, Loss: 0.17303746938705444
Epoch 5, Loss: 0.17300765216350555
Epoch 6, Loss: 0.17299066483974457
Epoch 7, Loss: 0.17297974228858948
Epoch 8, Loss: 0.17297209799289703
Epoch 9, Loss: 0.17296643555164337
Epoch 10, Loss: 0.17296205461025238
Evaluation on Original Dataset:
Accuracy: 1.00
Recall: 0.00
F1-Score: 0.00
```

In [47]:

```
# SMOTE Dataset
model_smote = MLP(input_size)
optimizer_smote = optim.Adam(model_smote.parameters(), lr=0.001)

print("\nTraining on SMOTE Dataset:")
train_model(X_train_smote, y_train_smote, model_smote, criterion, optimizer_smote, epochs=10)
print("Evaluation on SMOTE Dataset:")
evaluate_model(X_test, y_test, model_smote)
```

```
Training on SMOTE Dataset:
Epoch 1, Loss: 40.890220642089844
Epoch 2, Loss: 49.98793029785156
Epoch 3, Loss: 50.00005340576172
Epoch 4, Loss: 50.00003433227539
Epoch 5, Loss: 50.00002670288086
Epoch 6, Loss: 50.000022888183594
Epoch 7, Loss: 50.00001907348633
Epoch 8, Loss: 50.00001907348633
Epoch 9, Loss: 50.0000114440918
Epoch 10, Loss: 50.00000762939453
Evaluation on SMOTE Dataset:
Accuracy: 1.00
Recall: 0.00
F1-Score: 0.00
```

In [49]:

```
# Undersampled Dataset
model_under = MLP(input_size)
optimizer_under = optim.Adam(model_under.parameters(), lr=0.001)

print("\nTraining on Undersampled Dataset:")
train_model(X_train_under, y_train_under, model_under, criterion, optimizer_under, epochs=10)
print("Evaluation on Undersampled Dataset:")
evaluate_model(X_test, y_test, model_under)
```

```
Training on Undersampled Dataset:
Epoch 1, Loss: 50.0
Epoch 2, Loss: 50.0
Epoch 3, Loss: 50.0
Epoch 4, Loss: 50.0
Epoch 5, Loss: 50.0
Epoch 6, Loss: 50.0
Epoch 7, Loss: 50.0
Epoch 8, Loss: 50.0
Epoch 9, Loss: 50.0
Epoch 10, Loss: 50.0
Evaluation on Undersampled Dataset:
Accuracy: 1.00
Recall: 0.00
F1-Score: 0.00
```

In [50]:

```
print('Original Dataset: Likely higher precision but lower recall due to imbalance')
print('SMOTE Dataset: Expected to improve recall but may slightly affect precision due to synthetic samples.')
print("Undersampled Dataset: May improve precision but reduce recall due to loss of majority class data.")
```

```
Original Dataset: Likely higher precision but lower recall due to imbalance
SMOTE Dataset: Expected to improve recall but may slightly affect precision due to synthetic samples.
Undersampled Dataset: May improve precision but reduce recall due to loss of majority class data.
```

In [ ]: